

PPA 1

Title

Substrate Computation via Spatially Embedded Logic

Field of the Invention

This invention relates to computation, logic systems, artificial intelligence, and information processing, and more specifically to computational systems in which logic is embedded in spatial structure rather than executed symbolically or sequentially.

Background

All known computational systems are founded on symbolic abstraction, sequential evaluation, or externally defined rules. Even systems described as parallel, neural, or distributed rely on predefined representations, evaluation mechanisms, and objective criteria.

Such systems cannot natively generate or discover new structural logics. They operate within representational constraints imposed by their designers.

Biological systems demonstrate a different mode of computation in which structure itself participates directly in logic formation. Existing computational paradigms cannot faithfully reproduce this behavior.

Summary of the Invention

The invention discloses a class of computational systems in which:

- Logical operations are embedded directly in spatial or volumetric structure
- States are defined relationally rather than symbolically
- Computation occurs through intrinsic transformation rather than instruction execution
- Evaluation is not explicit, numerical, or symbolic

In such systems, logic is not applied to structure; logic *is* structure.

Detailed Description

The system consists of a finite or bounded substrate composed of spatially related states (“volumetric points”), where:

- Each state has intrinsic adjacency, symmetry, and transformability

- No global coordinate system is required
- No symbolic encoding of state is necessary
- No linear instruction sequence governs transformation

Transformations occur according to constraints intrinsic to the substrate's topology and symmetry relations.

Boolean completeness may emerge geometrically but is not executed symbolically.

Key Novel Claims (Provisional)

- Computation without symbols, instructions, or sequential evaluation
- Logic arising from spatial constraint rather than symbolic rule sets
- Systems in which transformation replaces evaluation
- Substrates in which structure directly generates structure